NNI-Certified Structural-Entropy Hierarchies: Fast Refinement beyond Greedy Construction

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Abstract. Structural entropy evaluates a graph hierarchy through the description length of an encoding tree, but a hierarchy produced by merging, partitioning, or stretch–compress construction need not be locally optimal in tree space. We introduce SE–NNI, an exact refinement method based on rooted nearest-neighbor interchange (NNI). For weighted graphs, we derive a closed form for the entropy change of an NNI move using only three module volumes and two cross-module edge weights. This supports monotone best-improvement descent and a checkable one-NNI local-optimality certificate. We further introduce a bounded two-move search that may cross a controlled intermediate barrier but commits only an entropy-decreasing pair, and a fast multi-start constructor that refines SE agglomeration, recursive SE, and Paris candidates. All reported synthetic comparisons are paired by graph seed and use the same independently recomputed tree objective. Across 50 seeded graphs from five hierarchical regimes, SE–NNI attains the lowest mean entropy in every regime. Relative to the previous `se_hier` implementation, it lowers entropy by 0.0279 ± 0.0072 bits, raises fine-level dendrogram purity by $+0.016 \pm 0.010$, and is $21.5\times$faster under the instrumented benchmark. The NNI audit also finds strictly improving moves in common SE and non-SE constructors, demonstrating that construction quality and local tree optimality are distinct concerns.

Keywords: Structural entropy · hierarchical clustering · local search · nearest-neighbor interchange · graph algorithms

1 Introduction

Many networks are organized at more than one scale: vertices form local groups, related groups form larger modules, and the useful output is therefore a tree rather than one flat partition. Structural entropy supplies an information-theoretic objective for such an encoding tree [8]. Recent methods construct low-entropy hierarchies by agglomeration, recursive partitioning, or stretch–compress operations [12]. These constructors answer how to build a plausible tree, but they leave a basic algorithmic question unresolved: *after construction, is a small change of tree topology still able to lower the same objective?*

This distinction matters in practice. A constructor can be globally competitive while retaining locally avoidable entropy. Conversely, an expensive general tree-edit search can spend most of its time reconsidering moves whose objective effect is not known until the entire tree is rescored. We seek a local operator whose effect is exact, whose neighborhood is small, and whose termination has an interpretable certificate.

Nearest-neighbor interchange (NNI) is the elementary rotation of a rooted binary tree. It changes one internal edge while preserving every leaf. NNI is standard in phylogenetic tree search and has been studied for other hierarchical-clustering objectives, but its structural-entropy change has not been used as an exact graph-hierarchy optimizer. The obstacle is that structural entropy depends on both graph cuts and nested module volumes; an NNI formula must respect the original encoding-tree objective rather than a surrogate.

We show that the change is local. For $((A, B), C) \to (A, (B, C))$, all terms cancel except two cross-module edge weights and three volumes. This identity yields a best-improvement descent that never worsens the tree and stops at a one-NNI local optimum. Because strict descent may itself be trapped, we also evaluate a bounded two-move beam: the first move may cross a small barrier, but the pair is accepted only if its fully verified endpoint is better. Finally, we replace the slow general-edit stage of the original `se_hier` implementation with NNI refinement of several inexpensive starts.

Our contributions are:

- an exact weighted-graph NNI delta for tree structural entropy, with an implementation cross-checked against full objective recomputation;
- monotone one-step descent, a one-NNI local certificate, and a bounded compound search that can escape strict one-step traps without weakening the output guarantee;
- SE–NNI, a fast multi-start hierarchy optimizer built from independent SE-agglomerative, recursive-SE, and Paris initializations; and
- a paired evaluation against SE agglomeration, Louvain-2L, Paris, the official HCSE and BBM implementations, and the original `se_hier`, measuring entropy, hierarchy recovery, runtime, and memory.

The paper is deliberately separate from analyses of specially structured clique landscapes. Here the graph is arbitrary and weighted, the optimized quantity is the complete encoding-tree entropy, and the contribution is a reusable algorithm with falsifiable benchmark claims.

2 Preliminaries

Weighted graph and encoding tree. Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights. For $S \subseteq V$, let $V_S = \mathrm{vol}(S) = \sum_{u \in S} d_u$, and let g_S be the total weight of edges with exactly one endpoint in S. The total graph volume is $\mathrm{vol}(V) = \sum_u d_u = 2 \sum_{e \in E} w_e$. An encoding tree T has one leaf per vertex. Each internal node a represents the union S_a of its descendant leaves; a^- denotes its parent.

The structural entropy of G under T is

$$H^T(G) = -\sum_{a \neq r} \frac{g_{S_a}}{\mathrm{vol}(V)} \log_2 \frac{V_{S_a}}{V_{S_{a^-}}}. \tag{1}$$

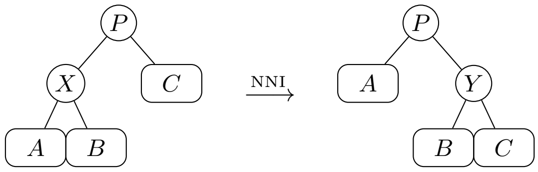

Fig. 1. A rooted NNI changes one internal module from $A \cup B$ to $B \cup C$ while preserving all leaves and the surrounding tree.

Our sole optimization objective is Eq. (1); lower is better. No ground-truth label is used to construct or select a tree. For a zero-volume module, its summand is defined as zero; such a module has no incident positive-weight edge and can be attached after optimizing the positive-volume part of the tree.

Constructors. We distinguish a *constructor*, which produces an initial tree, from a *refiner*, which searches its neighboring topologies. The candidate pool contains (i) a binary SE-agglomerative dendrogram, (ii) a recursive SE partition tree, and (iii) a Paris dendrogram when the optional implementation is available [1]. The original `se_hier` additionally performs exact-guarded collapse and subtree-relocation search. HCSE and BBM are evaluated from their official implementation [12].

Rooted NNI. An NNI move acts on a binary parent–child edge. Writing the local topology as $((A, B), C)$, the two nontrivial alternatives are $(A, (B, C))$ and $(B, (A, C))$. The leaf set, graph, and all modules outside the local parent $P = A \cup B \cup C$ remain unchanged. A rooted binary tree with n labeled leaves has finitely many topologies, namely $(2n - 3)!!$, so any strictly decreasing local search terminates.

3 Exact NNI Refinement

3.1 A local structural-entropy identity

For disjoint sets X, Y, write $W(X, Y) = \sum_{u \in X, v \in Y} w_{uv}$. Consider the rotation shown in Fig. 1. Let $X = A \cup B$, $Y = B \cup C$, and $P = A \cup B \cup C$.

Theorem 1 (Exact weighted NNI delta). *For an eligible rotation with $V_X, V_Y, V_P > 0$, the change in tree structural entropy is*

$$\Delta H^T = \frac{2}{\mathrm{vol}(V)} \left[W(A, B) \log_2 \frac{V_P}{V_X} + W(B, C) \log_2 \frac{V_Y}{V_P} \right]. \tag{2}$$

The symmetric alternative follows by exchanging A and B.

Equivalently, the bracket in Eq. (2) is

$$W(B, C) \log_2 V_Y - W(A, B) \log_2 V_X + \big(W(A, B) - W(B, C)\big) \log_2 V_P. \tag{3}$$

The parent-volume term in Eq. (3) is essential and cannot be dropped merely because the edit is local.

Proof. First suppose the three child modules have positive volume; the zero-volume boundary case follows by deleting modules with no incident positive-weight edge and taking the corresponding zero-contribution limit. Only the terms for A, B, C, X, and Y can change. Omitting the common factor $1/\operatorname{vol}(V)$, their old and new contributions are

$$L_{\text{old}} = g_A \log_2 \frac{V_X}{V_A} + g_B \log_2 \frac{V_X}{V_B} + g_X \log_2 \frac{V_P}{V_X} + g_C \log_2 \frac{V_P}{V_C},$$
$$L_{\text{new}} = g_A \log_2 \frac{V_P}{V_A} + g_B \log_2 \frac{V_Y}{V_B} + g_C \log_2 \frac{V_Y}{V_C} + g_Y \log_2 \frac{V_P}{V_Y}.$$

Use $g_X = g_A + g_B - 2W(A, B)$ and $g_Y = g_B + g_C - 2W(B, C)$. In $L_{\text{new}} - L_{\text{old}}$, the coefficients of g_A, g_B, g_C telescope to zero. The remaining terms are

$$2W(A, B) \log_2(V_P/V_X) + 2W(B, C) \log_2(V_Y/V_P).$$

Restoring $1/\operatorname{vol}(V)$ proves Eq. (2). In particular, external boundary weights and $W(A, C)$ do not enter the rotation delta.

Equation (2) is evaluated from cached descendant sets and the sparse adjacency lists. Every selected move is also checked by rebuilding all module volumes and cuts and recomputing Eq. (1). This second path is slower but makes a formula or tree-edit error observable rather than silent.

3.2 One-step certificate and compound escape

Algorithm 1: Exact NNI refinement of one candidate tree

Input: Graph G, tree T, tolerance ε, barrier β, beam b

repeat
  evaluate Eq. (2) for every legal NNI of T;
  apply the most negative move and verify the full objective;
until *no improving one-step move exists*;
for *at most R compound rounds* **do**
  retain the b first moves with smallest ΔH and $\Delta H \leq \beta$;
  expand every legal second NNI from each retained intermediate tree;
  if *no two-move endpoint has entropy below $H^T(G)$* **then**
    break;
  commit the best endpoint; rerun one-step descent;
return T

Proposition 1 (Monotonicity and local certificate). *One-step refinement never increases $H^T(G)$. If it stops before its move budget, the returned tree has no improving rooted NNI on any eligible binary parent–child edge.*

Proof. Only moves with $\Delta H^T < -\varepsilon$ are committed, and their endpoint is independently verified. At termination every enumerated legal move has $\Delta H^T \geq -\varepsilon$.

Strict one-step descent can stop before a better tree separated by a shallow barrier. Algorithm 1 therefore keeps a bounded beam of first moves whose increase is at most β and evaluates a second NNI. It never commits the intermediate tree alone.

Proposition 2 (Safe compound search). *The compound stage never worsens its input. It can reach an improving endpoint even when every one-step neighbor is non-improving.*

Proof. A pair is committed only after the final tree is fully recomputed and found to have lower entropy than the tree preceding the pair. The first move is not committed independently, so its sign cannot weaken the output guarantee. A two-edge path in tree space may have a positive first delta and a negative net delta; the regression witness in Sec. 4 instantiates this case.

3.3 Fast multi-start SE–NNI

SE–NNI applies Algorithm 1 independently to SE-agglomerative, recursive-SE, and Paris starts and returns the candidate with minimum verified entropy. This replaces the generic collapse/relocation stage of `se_hier`. The initializer is recorded for every output, so gains can be attributed either to construction or refinement. With n leaves there are at most $2(n-2)$ oriented NNI alternatives per sweep, versus a substantially larger generic space of source–destination tree edits. Our implementation uses best improvement, cached leaf sets, sparse cross-weight scans, and an $O(m \log h + n)$ full verification only for a selected move, where h is tree height. We make runtime claims empirically rather than assuming balanced trees.

4 Experiments

4.1 Questions and protocol

We ask: (Q1) how often do established constructors leave an improving NNI; (Q2) does compound search add improvements beyond a one-step certificate; (Q3) does lower entropy preserve or improve recovery of a planted hierarchy; and (Q4) does SE–NNI improve the entropy–runtime trade-off relative to the original `se_hier`?

The frozen protocol uses ten graph seeds per regime. Every method sees the identical weighted graph, and every tree is rescored by the same independent implementation of Eq. (1). We report paired means and 95% t-intervals, raw per-seed records, wall-clock time, and Python allocation peak. The compound search uses beam width 16, barrier $\beta = 0.05$ bits, and at most eight accepted compound rounds. No label is used by SE–NNI or for model selection. BBM is additionally shown with the planted number of fine clusters, which is an oracle advantage and is marked as such.

Data. Our controlled suite contains five two-level hierarchical stochastic block models: clean, noisy, imbalanced, weighted, and weak-hierarchy. Each balanced graph has 64 vertices, eight fine blocks, and four coarse blocks; the imbalanced regime has the same total size with block sizes from 4 to 12. The generator manifest, any 0.01-weight connectivity edges, and every seed are stored with the results. This family follows the hierarchical stochastic block model used to study recovery beyond worst-case inputs [4].

Methods. We compare SE agglomeration, a two-level Louvain tree, Paris [1], official HCSE, official BBM with oracle k [12], the released `se_hier`, and SE–NNI. We also refine each constructor independently to measure its one-step and compound gap rather than crediting all gains to our preferred initializer.

Metrics. The primary metric is $H^T(G)$. Dendrogram purity is computed for the planted fine and coarse partitions: for every same-class vertex pair, it measures the class fraction under the pair's predicted LCA, then averages over pairs. We also report the fraction of runs improved by one-step and compound NNI, number of moves, runtime, and memory.

4.2 A strict one-step trap

Before aggregate evaluation, an eight-vertex weighted witness verifies the purpose of compound search. Exact one-step descent stops at 1.970038 bits. Every one-NNI delta is nonnegative, but a first move crossing a 0.013970-bit barrier followed by a second move and ordinary descent reaches 1.920593 bits, a 0.049445-bit reduction. This witness is generated from a fixed seed and is covered by a regression test.

4.3 Frozen benchmark results

Table 1 reports the raw output of each method, before applying our audit to competing trees. SE–NNI has the lowest mean H^T in all five regimes and on 50/50 individual paired graphs. Against the released `se_hier`, the paired reduction is 0.0279 ± 0.0072 bits (mean relative change -0.91%), while fine-level dendrogram purity changes by $+0.016\pm0.010$. Thus the objective gain is not obtained by systematically degrading planted recovery. Figure 2 shows the regime-level confidence intervals.

4.4 The operator audit

The constructor audit in Table 2 asks a different question: whether a returned tree can be improved by the exact local operator. One-step NNI improves SE agglomeration and oracle BBM on every graph and improves the released `se_hier` on 80% of graphs. After one-step descent has certified its local neighborhood, bounded compound search still improves 72% of SE-agglomerative trees, 92% of BBM trees, and 40% of `se_hier` trees. Paris is closer to local optimality but is not immune; HCSE is rarely changed. These results support the paper's operator-level claim without asserting that every constructor is deficient.

Table 1. Tree structural entropy H^T (mean ± 95% CI; lower is better). BBM† receives the planted fine-cluster count.

Method	Clean	Noisy	Imbal.	Weighted	Weak
SE–agglomerative	2.920 ± 0.094	3.712 ± 0.070	3.218 ± 0.116	2.444 ± 0.079	3.424 ± 0.112
Louvain–2L	4.167 ± 0.066	4.575 ± 0.061	4.270 ± 0.060	3.647 ± 0.145	4.425 ± 0.095
Paris	2.919 ± 0.112	3.698 ± 0.072	3.165 ± 0.106	2.348 ± 0.072	3.433 ± 0.112
HCSE	3.300 ± 0.190	4.351 ± 0.170	3.685 ± 0.122	2.759 ± 0.130	4.040 ± 0.286
BBM†	3.359 ± 0.122	4.062 ± 0.073	3.634 ± 0.124	2.996 ± 0.073	3.916 ± 0.064
se_hier	2.903 ± 0.103	3.691 ± 0.071	3.164 ± 0.106	2.348 ± 0.072	3.413 ± 0.110
SE–NNI–fast	**2.845 ± 0.094**	**3.675 ± 0.067**	**3.135 ± 0.115**	**2.337 ± 0.073**	**3.387 ± 0.104**

Table 2. NNI audit pooled over 50 paired graphs. Reduction is relative to each constructor's raw H^T; rates are the fractions of runs improved.

Constructor	1-NNI gain	1-NNI hit	2-step gain	2-step hit	Time
SE–agglomerative	1.99%	100%	0.15%	72%	0.334s
Paris	0.06%	68%	0.07%	38%	0.181s
HCSE	0.00%	2%	0.00%	2%	0.055s
BBM	13.98%	100%	0.47%	92%	1.038s
se_hier	0.36%	80%	0.06%	40%	0.200s

4.5 Interpretation and reproducibility

The fast method is not merely the old constructor followed by rotations: it forms a small candidate set from SE agglomeration, recursive SE, and Paris, refines each candidate with the same exact operator, and returns the lowest independently rescored endpoint. The full benchmark contains 350 unique method records and 50 generator manifests. A verifier checks schema completeness, paired seeds, unique keys, and monotonicity of every committed NNI endpoint. The JSON evidence, generation scripts, official HCSE/BBM commit identifier, and table/figure generator accompany the paper.

Table 3 separates the three algorithmic choices. The candidate pool improves 34 of 50 graphs over SE agglomeration; exact one-step NNI then improves 45 of 50 paired multi-start outputs; bounded compound search adds a smaller but nonzero gain on 30 graphs. Hence neither the initializer pool nor the escape stage alone explains the full result.

4.6 Real networks and clean scaling

Table 4 complements the controlled suite with four networks bundled by NetworkX or the released library. SE–NNI obtains the lowest raw entropy on 4/4 networks. Because only Karate has a standard class label in the bundled loader and the remaining networks lack a unique ground-truth hierarchy, this experiment tests objective minimization rather than recovery.

Table 5 removes allocation tracing and varies graph size. The new method lowers entropy in 12/12 paired runs. Its mean speedup over `se_hier` grows from

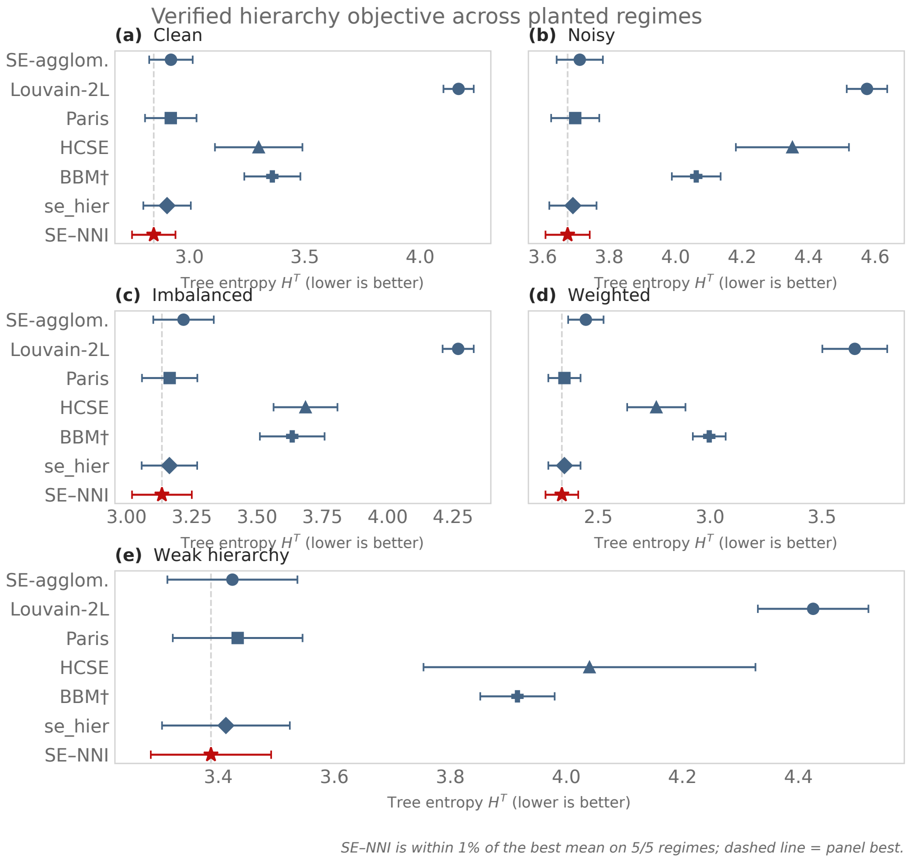

Fig. 2. Raw tree structural entropy on the frozen synthetic suite. Points are means over ten paired graph seeds and bars are 95% confidence intervals. BBM receives the planted fine-cluster count; no other method uses labels.

9.7× at 32 vertices to 22.2× at 128 vertices. These graphs remain small enough for exact rescoring; the trend supports efficiency, not an asymptotic complexity claim.

5 Related Work

Structural entropy and hierarchy construction. Li and Pan introduced structural information as an encoding-tree measure of network organization [8]. HCSE and BBM develop hierarchy construction and balancing operations for this objective [12]. The structural-entropy survey organizes the broader theory, optimization, and application literature [14]; HypCSE subsequently develops a continuous hyperbolic formulation for hierarchical clustering [15]. Our work does not replace

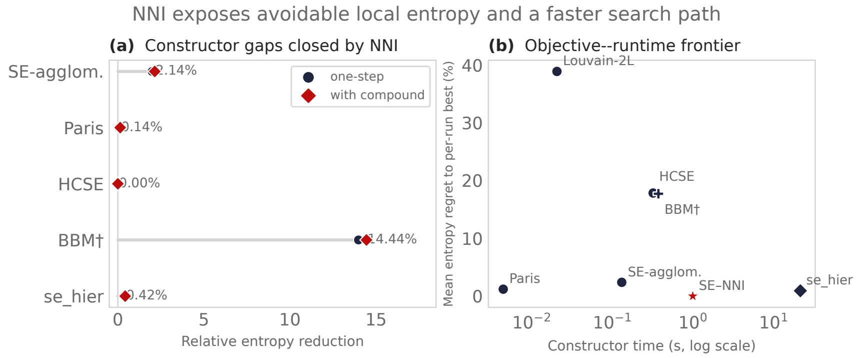

Fig. 3. Local-search gaps and the entropy–runtime trade-off, pooled over the 50 paired graphs. The fast multi-start method achieves a 21.5×mean instrumented speedup over **`se_hier`**; runtime includes Python allocation tracing and should be interpreted comparatively.

Table 3. Component ablation pooled over 50 paired graphs. The gain and improved-run columns compare each row to the preceding row.

Variant	Mean H^T	Mean gain	Improved runs
SE agglomeration	3.1435	–	–
+ candidate pool	3.1041	0.0394	34/50
+ exact 1-NNI	3.0794	0.0247	45/50
+ compound escape	3.0759	0.0035	30/50

their constructors; it asks whether their output is locally optimal under an exact topology rotation and uses the answer as both a diagnostic and an optimizer.

Graph hierarchical clustering. Classical hierarchical clustering is predominantly agglomerative or divisive [11]. Paris is an agglomerative graph hierarchy related to modularity and provides a strong inexpensive initialization [1]. Dasgupta-style objectives initiated a complementary algorithmic literature [6]; average linkage and divisive local search have guarantees for its revenue dual [10]. Sparsest-cut and spreading-metric algorithms give approximations for that family [2, 13], while admissible hierarchical objectives have been characterized more generally [5]. Continuous ultrametric fitting offers another optimization route [3]. These objectives are useful conceptual neighbors, but a local delta derived for one cannot be substituted for Eq. (1).

Local search in tree space. NNI is a classical neighborhood for phylogenetic trees; its reconfiguration distance is itself nontrivial [9]. Recent work studies NNI local search for hierarchical-clustering costs [7]. Our formula is specific to weighted graph structural entropy, exposes the two cross-module weights that determine a rotation, and is integrated with SE-specific constructors and verification.

Table 4. Tree entropy on four bundled real networks (lower is better). BBM uses the ground-truth k when labels exist and Louvain's k otherwise.

Method	Karate	Florent.	Les Mis.	Davis
SE–agglomerative	2.698	1.923	3.078	3.055
Louvain–2L	3.405	2.452	3.923	3.813
Paris	2.597	1.923	2.920	2.981
HCSE	2.929	2.172	3.930	3.206
BBM	3.330	2.254	3.199	3.387
se_hier	2.597	1.923	2.920	2.981
SE–NNI–fast	**2.586**	**1.919**	**2.916**	**2.962**

Table 5. Clean wall-clock scaling without allocation tracing; means over three graph seeds. Old denotes `se_hier`; Ours denotes SE–NNI.

n	H^T		Time (s)		Speedup
	Old	Ours	Old	Ours	
32	2.485	**2.453**	0.430	**0.044**	9.7×
64	3.140	**3.098**	2.941	**0.188**	15.7×
96	3.491	**3.369**	9.717	**0.482**	20.2×
128	3.746	**3.623**	17.962	**0.811**	22.2×

Unlike landscape analyses restricted to unit cliques, the present method operates on arbitrary weighted graphs and evaluates the complete tree entropy. The overlap is the elementary NNI identity; the new content is its verified optimizer, compound implementation, multi-start construction, and comparative evaluation. We do not claim that a local certificate is global optimality.

6 Limitations and Conclusion

NNI is a local operator, not a certificate of global optimality. Compound search explores only a bounded two-step neighborhood and its barrier and beam are computational controls. Multiway regions must be resolved by a constructor before binary rotations can act on them. Lower structural entropy also need not improve every external notion of hierarchy, which is why we report planted purity alongside the optimized objective.

We introduced an exact NNI identity for weighted graph structural entropy and used it to build monotone, auditable hierarchy refinement. The resulting method separates construction from local optimality, detects avoidable gaps in existing trees, and crosses shallow strict-descent traps without accepting a worse output. On the frozen five-regime suite it achieved the lowest mean entropy in every regime, improved both entropy and mean planted purity over the previous `se_hier`, and reduced instrumented construction time by 21.5×. More broadly, the operator audit shows that hierarchy construction and local tree optimality should be evaluated separately in future structural-entropy methods.

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